# Python Activity-Based Learning

## Case Study 12: Attendance Recovery Computation

### 1. Problem Statement

Calculate a student's current attendance percentage and determine the minimum consecutive classes the student must attend to reach a required attendance percentage.

### 2. Objectives

- Calculate attendance percentage using arithmetic operators.
- Use a loop to find the minimum future classes needed.
- Compare scenarios for a required threshold.

### 3. Assumptions

- Attendance percentage is attended classes divided by total classes, multiplied by 100.
- The default eligibility threshold is 75%.
- Future classes are all attended; a student cannot attend more classes than the total classes held.

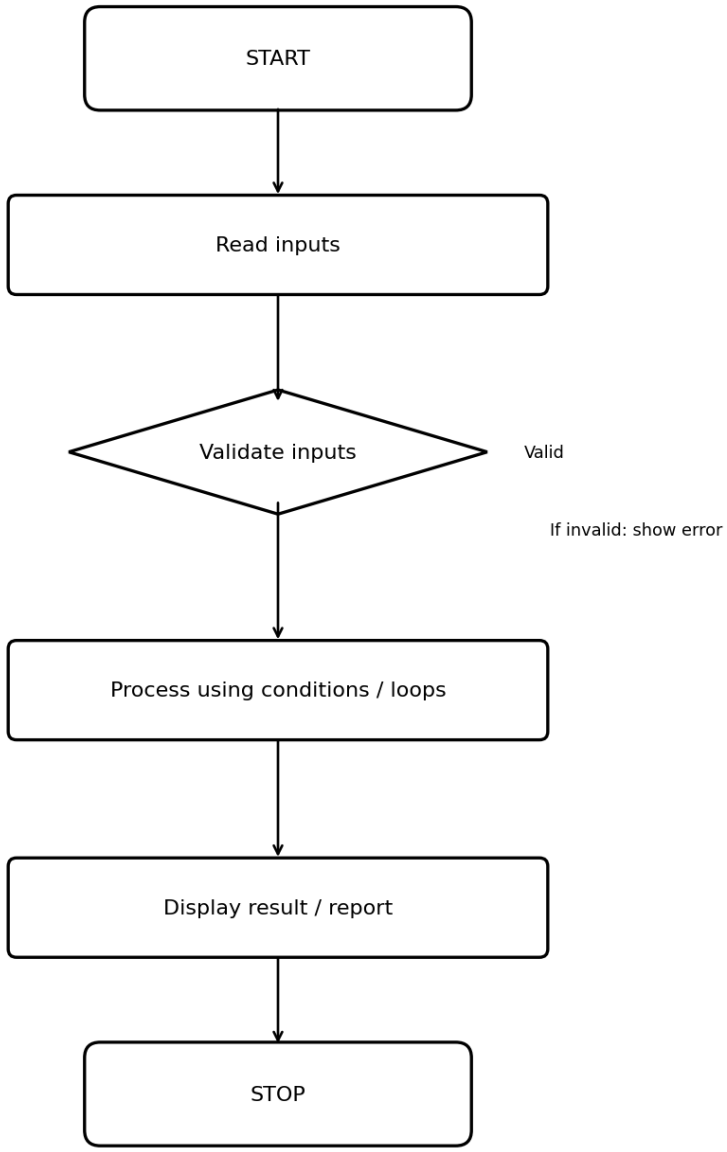
### 4. Algorithm

1. Read classes held and classes attended.
2. Validate that values are sensible.
3. Calculate current attendance percentage.
4. If attendance is below the threshold, repeatedly add one attended class and one total class until the threshold is reached.
5. Display the current percentage and required consecutive classes.

### 5. Flowchart

The diagram below shows the overall program flow. The validation decision represents the input checks described in the algorithm.

**Flowchart: Attendance Recovery Computation**




## 6. Pseudocode

```
START
READ total_classes, attended_classes, threshold
IF total_classes <= 0 OR attended_classes < 0 OR attended_classes > total_classes THEN DISPLAY invalid
input
ELSE
    current = attended_classes / total_classes * 100
    future = 0
    WHILE (attended_classes + future) / (total_classes + future) * 100 < threshold
        future = future + 1
    END WHILE
    DISPLAY current and future
END IF
STOP
```

## 7. Python Source Code

The solution uses only basic Python features from the assigned syllabus.

```
total = int(input("Total classes held: "))
attended = int(input("Classes attended: "))
threshold = float(input("Required attendance percentage: "))

if total <= 0 or attended < 0 or attended > total or threshold <= 0 or threshold > 100:
    print("Invalid input.")
else:
    current = attended / total * 100
    future = 0
    while (attended + future) / (total + future) * 100 < threshold:
        future += 1

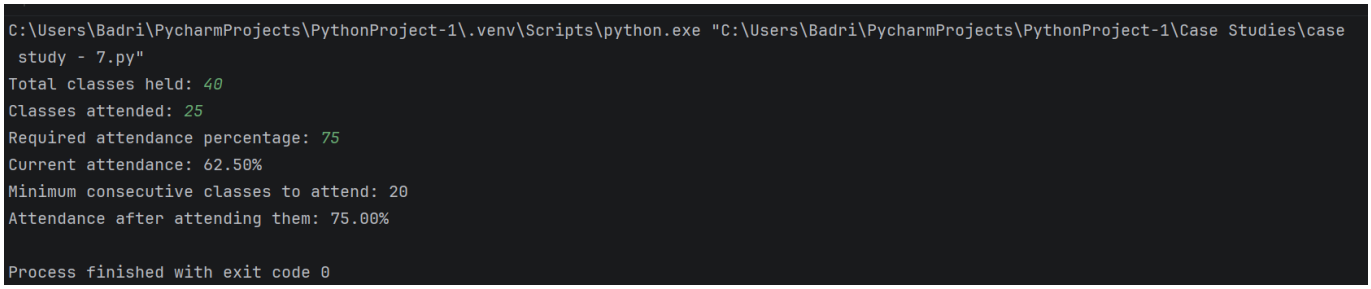
    print(f"Current attendance: {current:.2f}%")
    if future == 0:
        print("You already meet the attendance requirement.")
    else:
        print(f"Minimum consecutive classes to attend: {future}")
        final_percentage = (attended + future) / (total + future) * 100
        print(f"Attendance after attending them: {final_percentage:.2f}%")
```

## 8. Test Cases

| Test input / condition | Expected result | Purpose |
|---|---|---|
| 40 held, 25 attended, threshold 75 | 62.5% | Minimum future classes calculated by loop |
| 40 held, 32 attended, threshold 75 | 80% | Already eligible; 0 classes required |
| 10 held, 11 attended | Invalid record | Invalid input message |

## 9. Sample Output Screenshot

Sample console output for one test run:




## 10. Conclusion

The program uses a while loop to find the smallest number of future classes needed to reach an attendance target.

Note: Assumed rates, thresholds, and sample values are for this educational exercise and can be changed if the instructor supplies different rules.